



Big Data Visual Analytics (CS 661)

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Study Materials for Lecture 11

- Visualizing High-Dimensional Data: Advances in the Past Decade; S. Liu et al., TVCG2016
- t-SNE: <https://distill.pub/2016/misread-tsne/>
- UMAP: <https://pair-code.github.io/understanding-umap/>

Acknowledgements

- Some of the following slides are adapted from the excellent course materials and tutorials made available by:
 - Prof. Klaus Mueller (State University of New York at Stony Brook)
 - Prof. Tamara Munzner (University of British Columbia)
 - Zaur Fataliyev (Research Scientist – Meta)
 - Andreas Kollegger (neo4j)

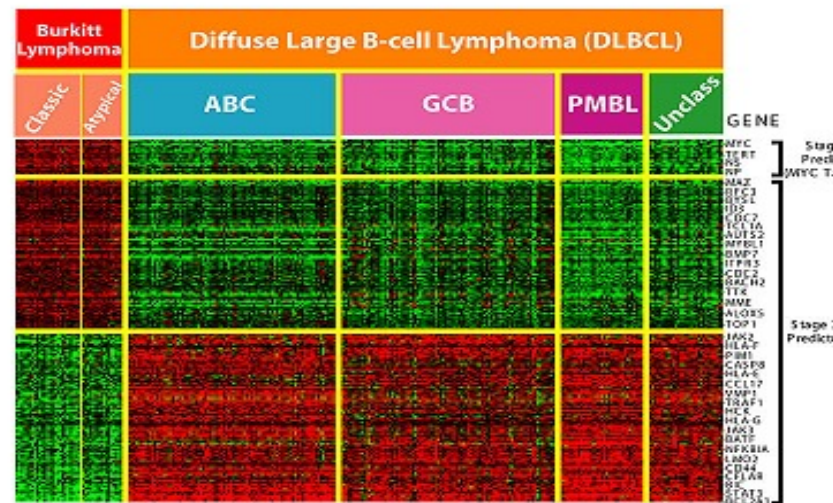
High Dimensional Data

- In statistics, high dimensional data is a data where the number of attributes (features) are larger than the number of samples
- In practice, often, when a data set has large number of attributes, it is also referred to as high dimensional data
 - Examples: biological data, gene expression data, social media user data, network data, etc.

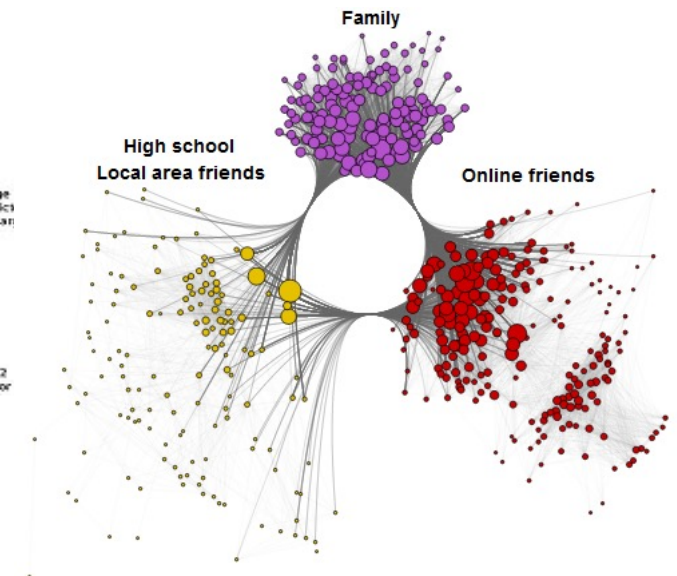
Dataset	Features	GP	DT	NB	KNNs	SVMs	RF
Adenocarcinomas(58)	50	99.83±0.009	73.33±1.6	87.66±3	86±1.3	91.33±0.7	77.67±1.5
	100	98.46±0.08	76.33±3.3	90±3.3	93.3±1.0	91.67±1.2	89.67 ± 1.5
	150	97.14±0.9	86±2.7	90±3.3	91.67±1.3	95±1.6	89.67±1.5
Oral Mucosa(79)	50	99.95±0.002	79.82±1.8	74.82±3.44	57.14±6.6	81±1.4	78.3±2.2
	100	98.57±0.08	70.89±0.16	69.46±3.8	62.14±2.1	77.32±2.6	81.25±5.9
	150	96.93±0.17	68.39±0.95	65.71±2.7	64.64±1.3	81.75±1.4	78.75±6.7
B-Cells(79)	50	99.41±0.03	77.32±2.6	74.82±3.44	82.32±2.2	85±4.7	82±5.5
	100	97.28±0.15	72.32±4.2	76.07 ±3.0	77.57±2.2	88.75±3.5	83.75±5.1
	150	96.59±0.19	71.25±9	78.57±2.2	76.25±6.7	87.5±3.9	83.75±5.1
Placenta(76)	50	99.91±0.005	77.49±2.5	68.75±9.8	70.71±4.7	84.28 ± 0.45	84.28±0.45
	100	99.30±0.03	73.39±5.1	67.5±10.7	69.46±5.1	84.28±0.45	84.28±0.45
	150	97.95±0.11	70.71±4.2	68.75±9.8	69.46±5.1	84.28±0.45	81.6±1.2
Melanoma(83)	50	97.64±0.13	88.19±4.1	94.16±1.8	95.13±2.4	91.67±1.3	96.52±1.0
	100	97.06±0.16	84.3±2.9	92.77±1.67	95.13±1.53	96.3±2.8	97.77±0.7
	150	96.37±0.51	85.5±3.3	95.2±1.4	96.3±1.14	97.63±0.7	97.77±0.7
Breast cancer(97)	50	97.87±0.12	52.77±0.8	49.44±1.9	43.22±0.3	54.66±0.2	56±0.14
	100	96.75±0.18	52.55±2.5	49.33±1.9	47.22±1.1	51.55±1.2	55.88±0.1
	150	96.90±0.17	51.67±2.2	51.44±1.3	52.1±2.4	52.55±0.9	57±0.45
Skeletal Muscle(110)	50	99.24±0.04	65.45±2.2	69.09±7.4	74.54±0.57	89.09±2.2	83.63±0.5
	100	98.69±0.07	71.81±5.4	66.36±2.0	83.63±5.1	98.18±0.57	82.72±0.28
	150	98.27 ±0.09	63.36±0.8	68.18±1.43	85.45±1.72	97.27±2.0	81.81±2.8
Osteoarthritis(139)	50	99.90±0.005	71.97±1.5	72.74±3.7	78.4±0.46	86.97±3.1	78.46±1.9
	100	99.23±0.04	70±9.4	69.12±2.4	84.23±2.5	90.65±0.52	81.31±1.04
	150	98.73±0.07	82.03±0.8	67.69±2.9	83.51±2.7	94.23±0.68	77.74±2.17

<https://doi.org/10.1371/journal.pone.0196385.t004>

Tabular data



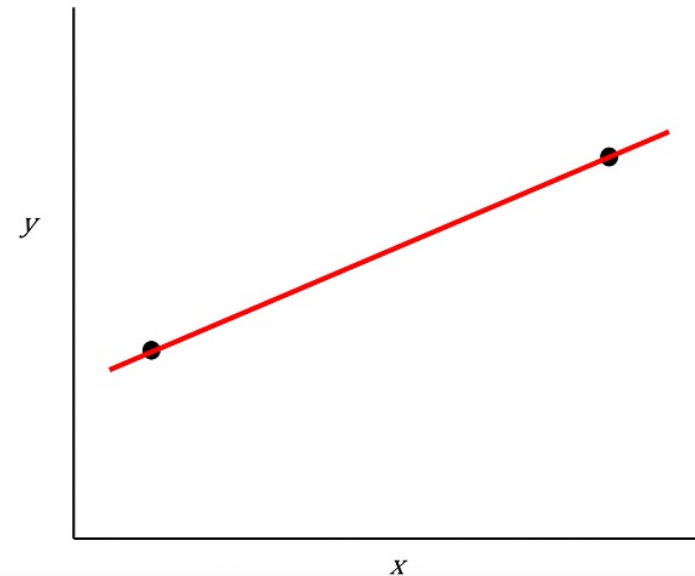
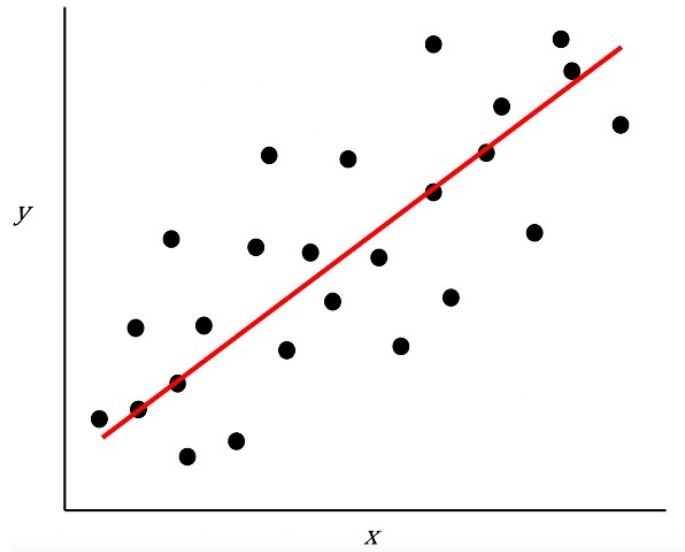
Microarray data



Graph data

Intuition: Why High-Dimensional Data Can Be A Problem?

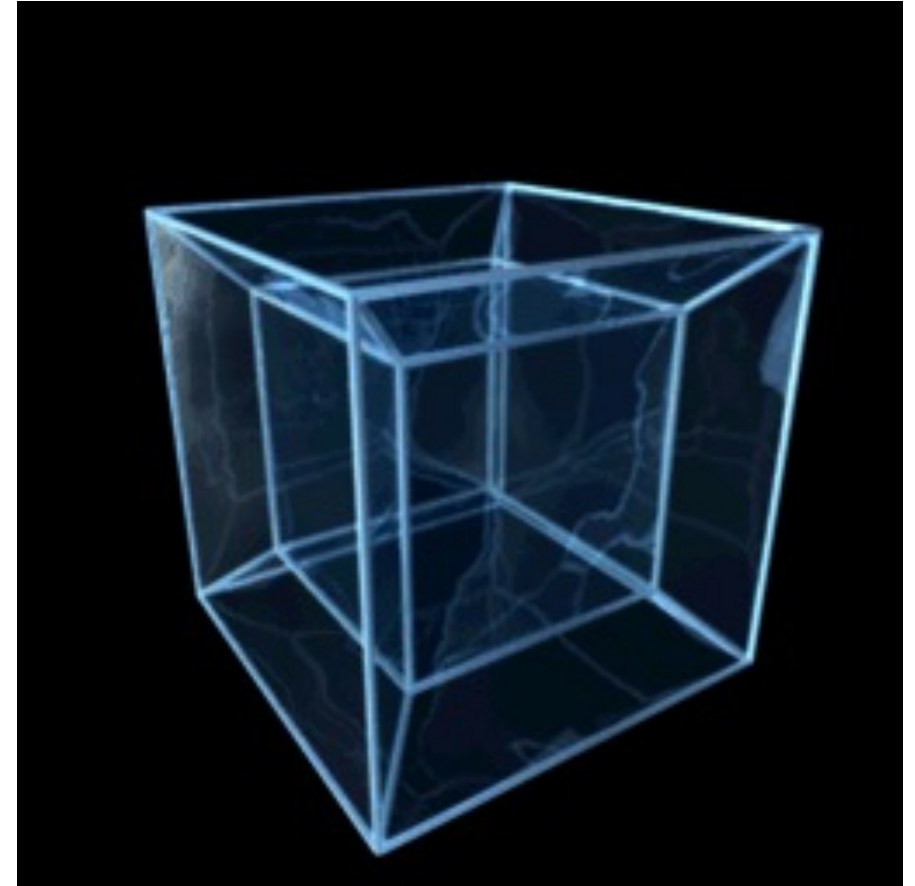
- Imagine a situation in which the number of observations and features in a dataset are almost equal
- Effective number of observations per features is low
- Result: Models (Statistical or ML) can overfit and so less generalizable



High dim. data,
overfitting in regression

Understanding High-Dimensional Objects

- Feature vectors are typically high dimensional
- We do not understand such vectors well – why?
- Because we don't learn to see/perceive high-dim objects when our vision system develops
- We only perceive 3D world!



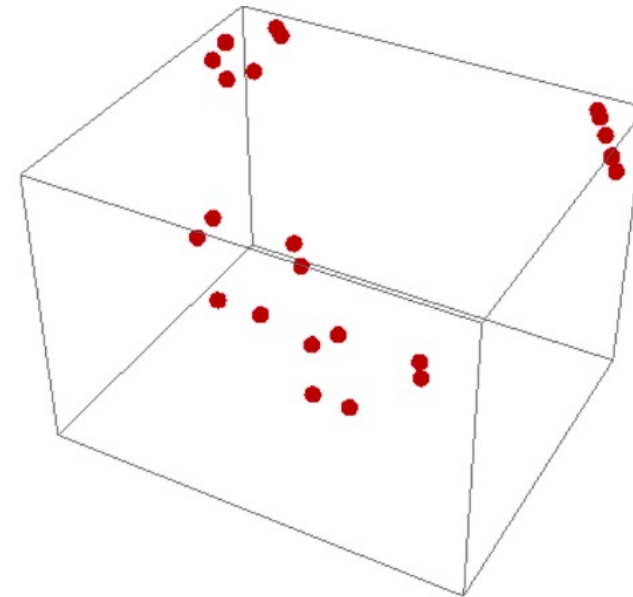
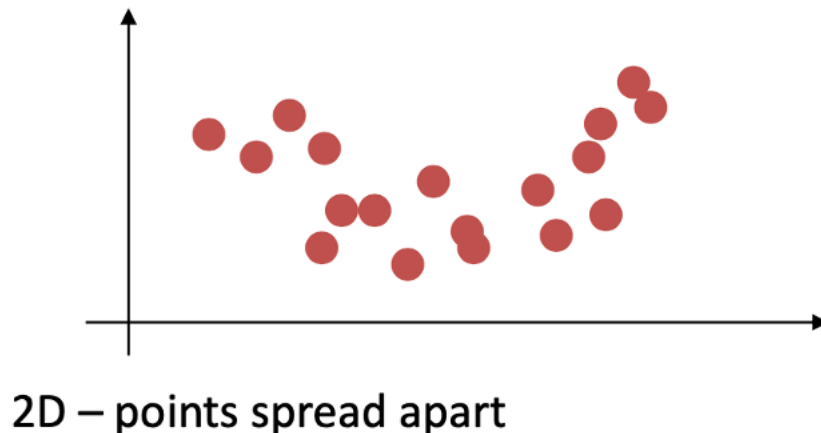
3D projection of a 4D cube

Curse of Dimensionality

- A phenomenon related with high dimensional data
 - Challenging to identify meaningful patterns while analyzing and visualizing the data
- With increasing dimensionality, the volume of the space increases rapidly, making the data sparse in high dimension
- To obtain a reliable result, the amount of data needed often grows exponentially with the dimensionality
- Distance computation between objects in high dimensional space becomes difficult

Sparseness in High Dimensional Space

- Space gets extremely sparse
 - with every extra dimension points get pulled apart further
 - distances become meaningless



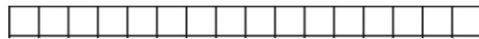
4D, 5D, ... – sparseness grows further

Space and Memory Management

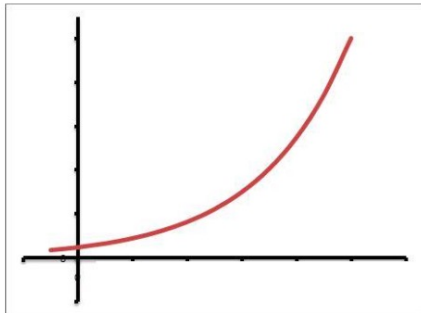
- Indexing (and storage) gets very expensive with increasing dimensionality
 - Exponential growth in the number of dimensions

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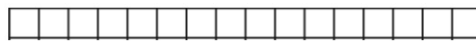


16 cells

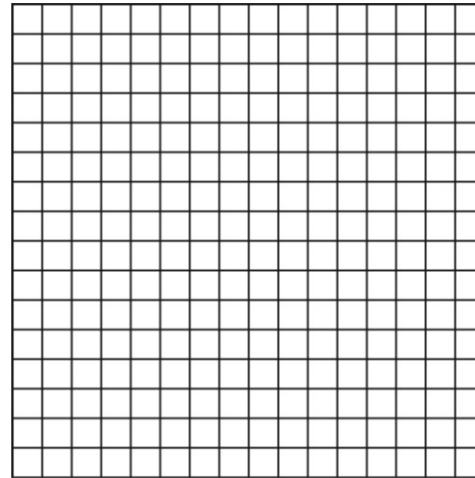
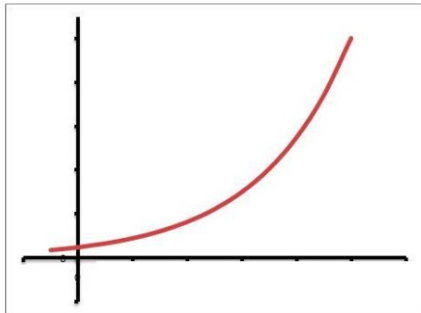


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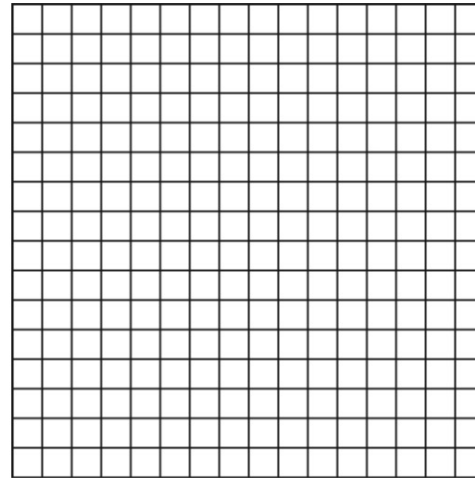
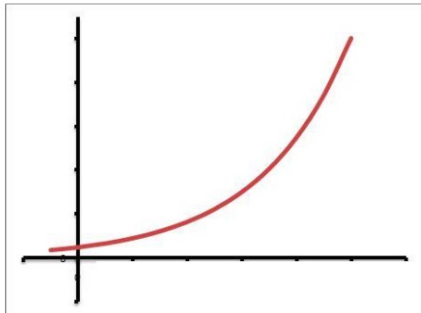
$16^2 = 256$ cells

Space and Memory Management

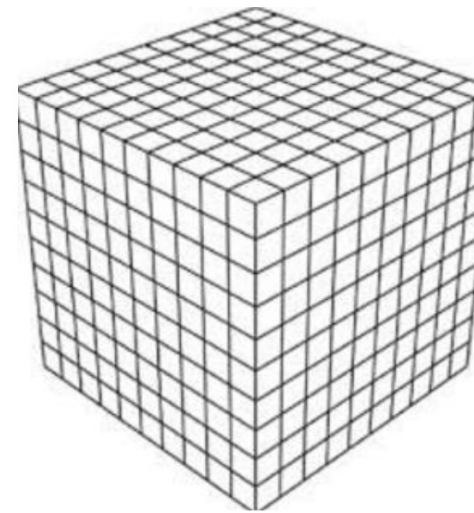
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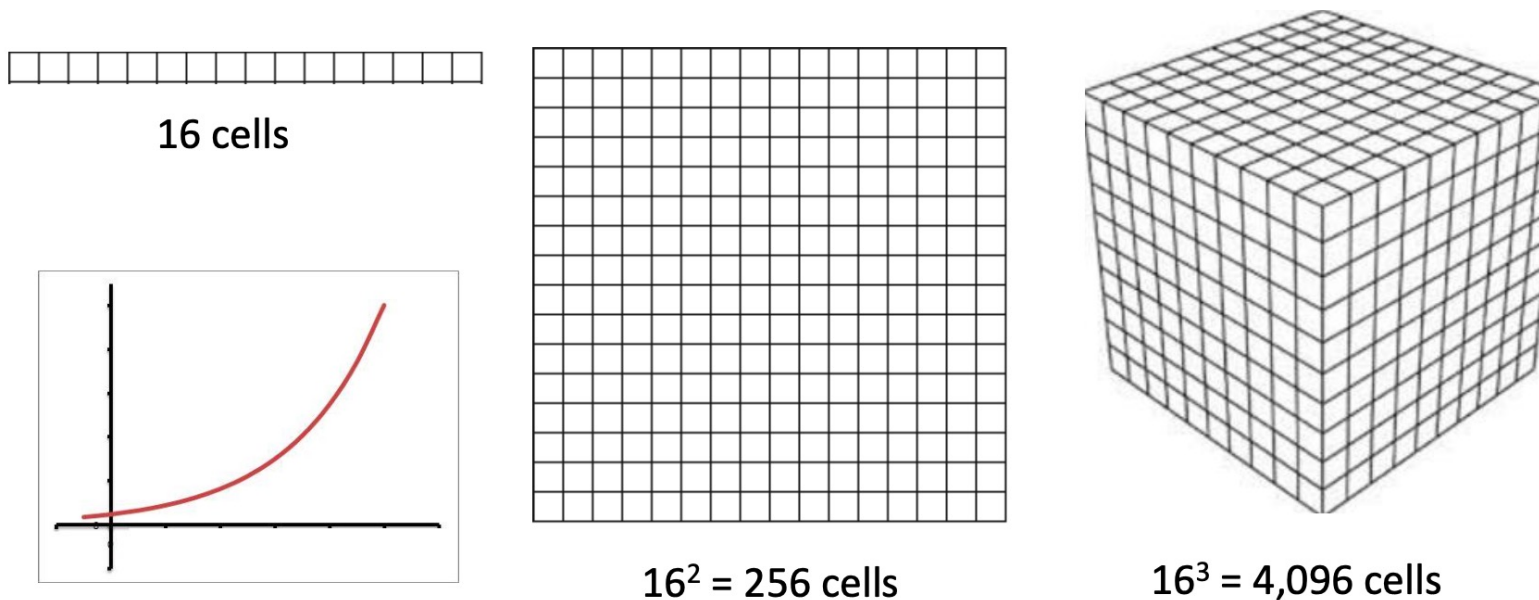
$16^2 = 256$ cells



$16^3 = 4,096$ cells

Space and Memory Management

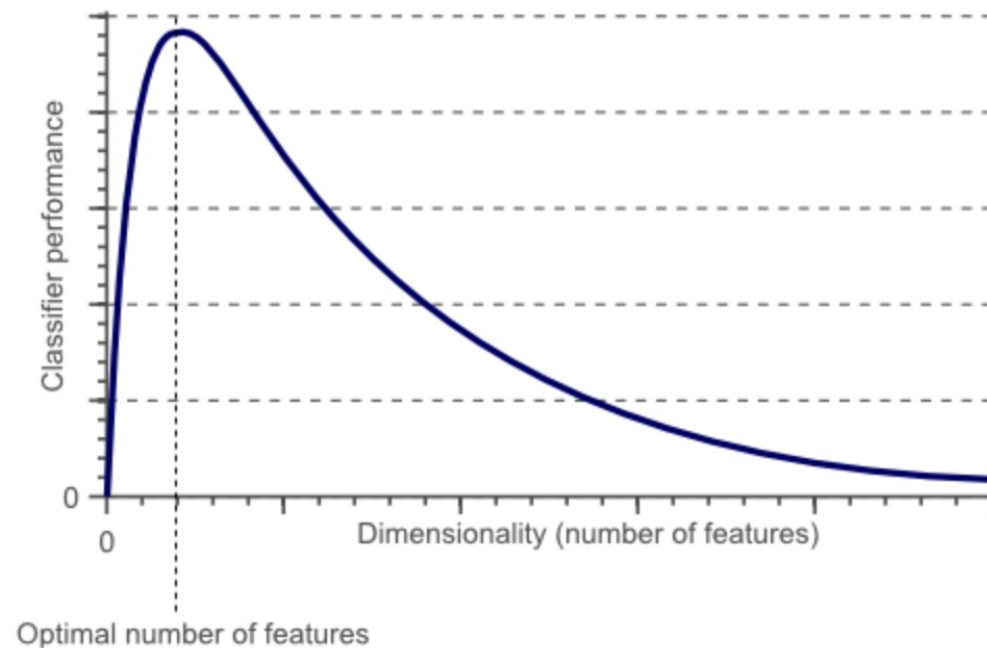
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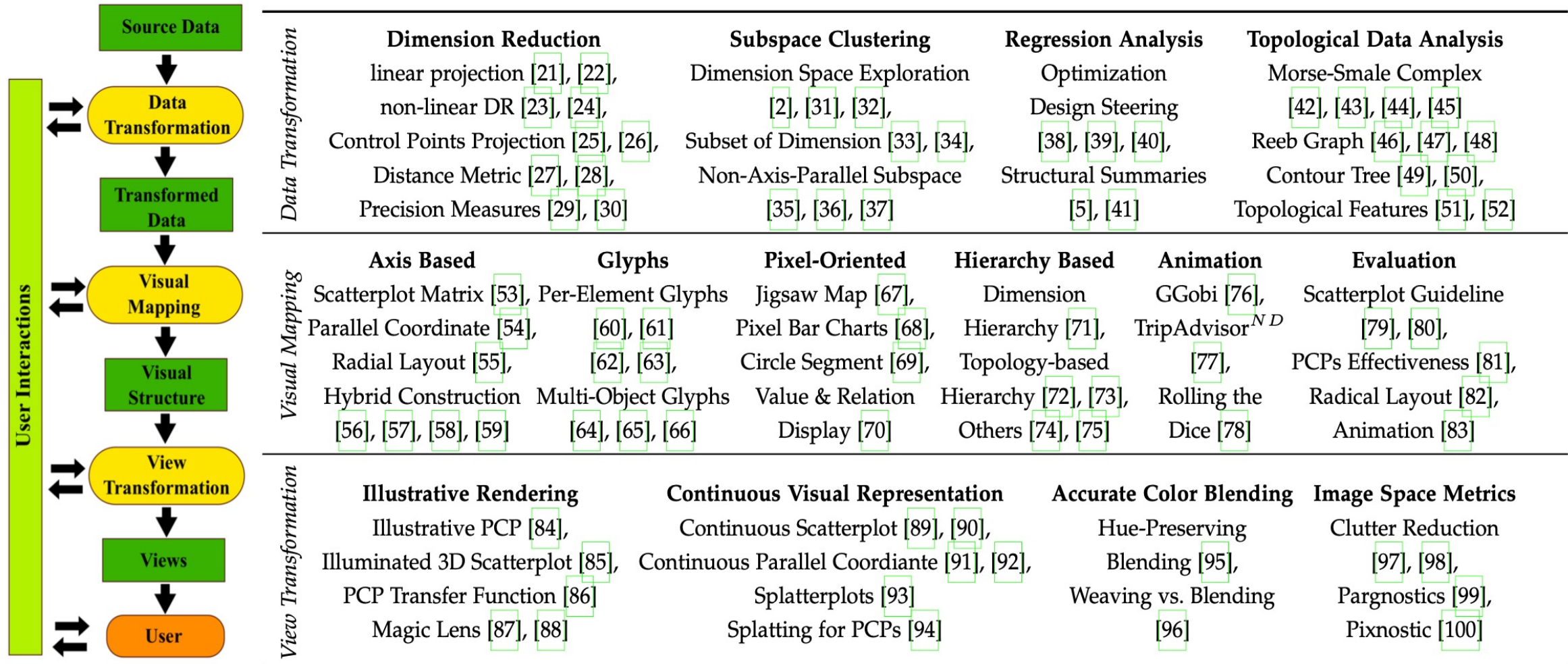
- 4D: 65k cells 5D: 1M cells 6D: 16M cells 7D: 268M cells

High Dimension: In Machine Learning

- Hughes' Phenomenon: With a fixed number of training samples, the average (expected) predictive power of a classifier or regressor first increases as the number of dimensions or features used is increased but beyond a certain dimensionality it starts deteriorating instead of improving steadily



High Dimensional Data Analysis and Visualization



Visualizing High-Dimensional Data: Advances in the Past Decade

Dimensionality Reduction: Why?

- Produce embedding of high dimensional data into low dimensional space
- Visual analysis of high dimensional data
- Useful for feature engineering in ML techniques
- Helps in finding redundant features from large scale data